

AquaPerma-NorthMB¹:
Detecting Surface Water Changes in Northern Manitoba
Using Multi-Temporal Satellite Imagery
Final Paper
Domenica Rosina Burroughs – 3053068
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Dr. Liu
University of Winnipeg

¹ Github Repo: <https://github.com/domierosina/AquaPerma-NorthMB>

1 Introduction

Northern Manitoba is undergoing rapid hydrological and ecological change due to shifting climate conditions. Rising temperatures, increased extreme precipitation, and altered seasonal freeze–thaw cycles are transforming surface water patterns, particularly in discontinuous permafrost regions. These landscapes contain dense networks of lakes, wetlands, and rivers, which are sensitive to permafrost thaw, snowmelt, and topography [1, 2, 3, 4]. Degrading permafrost increases surface ponding, alters vegetation, and can destabilize critical infrastructure such as the Hudson Bay Railway [3, 4]. These processes highlight the need for improved monitoring of surface water dynamics to inform adaptation planning.

Remote sensing provides an efficient method to detect surface water and quantify changes over time. Spectral water indices, particularly the Normalized Difference Water Index (NDWI), enable robust detection of open water using visible and near-infrared reflectance [5, 6, 7, 8]. In high-latitude environments, satellite-derived water masks effectively track long-term hydrological change and identify climate-driven variations in ponds, lakes, and wetlands [5, 8]. Recent studies highlight multi-decadal shifts in water extent, including increased thermokarst ponding and decreases in lake area due to drainage or evaporation [8, 18].

Despite these advancements, accessible workflows for northern hydrological monitoring remain limited. Cloud-based platforms like Google Earth Engine require stable internet and substantial computing resources, while local processing often demands optimized workflows and careful resource management.

1.1 Statement of the Problem and Rationale

Northern Manitoba lacks lightweight, accessible workflows for analyzing multi-temporal satellite imagery to monitor surface water changes. Local governments, communities, and

researchers benefit from tools that run on modest hardware without compromising analytical accuracy. This project addresses this gap by developing a reproducible, open-source processing pipeline for detecting surface water changes over five years using Landsat and Sentinel-2 imagery, supporting climate adaptation, land-use planning, and co-management with Indigenous communities [1, 2, 9].

1.2 Literature Review

Northern high-latitude regions experience some of the fastest climate-driven environmental changes globally. Permafrost thaw alters soil structure, subsurface hydrology, and thermal regimes, leading to complex patterns of surface water expansion and contraction [3–5, 8]. Increases in thermokarst ponds occur where ground ice melt causes subsidence, while some lakes shrink due to enhanced drainage and changes in the water balance. These diverse responses reflect interactions among permafrost type, soil ice content, watershed connectivity, and geomorphic setting.

Remote sensing is central for tracking these transitions, enabling repeated observation of large and often inaccessible areas. Optical satellite data from Landsat, Sentinel-2, and MODIS effectively map lakes, wetlands, and inundation at multiple scales. NDWI and MNDWI distinguish open water from vegetation, soil, and shadows, supporting assessment of lake change, thermokarst development, and hydrological impacts of permafrost thaw, wildfire, and climatic extremes [5, 8, 18].

Practical barriers remain, such as cloud-based workflows require broadband and can be constrained by quotas, and local workflows demand computational resources, coding expertise, or specialized software. Transparent, reproducible, and resource-efficient workflows are needed for northern communities, smaller research teams, and Indigenous organizations. Given the

pressing need for climate adaptation planning in northern Manitoba, workflows using freely available satellite imagery with minimal computational requirements can broaden access to environmental monitoring and promote collaborative, co-management initiatives.

2 Hypothesis and Research Aims

2.1 Hypothesis

A reproducible, open-source digital image processing pipeline using multi-temporal satellite imagery can effectively detect, quantify, and visualize surface water changes in Northern Manitoba through NDWI-based analyses and temporal differencing.

2.2 Research Aims

- Develop a local, resource-efficient workflow for detecting surface water from Landsat DSWE and Sentinel-2 imagery [9, 10].
- Generate NDWI-based water masks across multiple years to capture spatial and temporal variability [8].
- Quantify water changes over five years using area calculations and temporal differencing.
- Produce visualizations and statistics illustrating hydrological patterns and climate-related trends.

3 Materials and Methodology

3.1 Study Area

The final Area of Interest (AOI) covers 69.23 km² of northern Manitoba, selected from an original ~650,000 km² to balance computational feasibility and landscape representation [18]. This subset includes lakes, wetlands, and rivers typical of sub-Arctic environments. Limiting the AOI enabled reliable cloud masking, NDWI-based water detection, and temporal analysis of multi-temporal satellite imagery without compromising hydrological representativeness.

3.2 Data Sources and Sensor Justification

Sentinel-2 Level-2A and Landsat 8/9 Collection 2 Level-2 imagery (2016–2021) were used for surface water analysis. AOIs were set in Google Earth Engine and exported as KML files. Imagery was accessed via the openEO API (VITO/Terrascope) as spatiotemporal data cubes, filtered spatially to the AOI, temporally to summer months (June–August), and by cloud coverage (<30%) [9] – [12]. Annual composites were generated through temporal reduction and exported as multi-band GeoTIFFs for local NDWI computation.

A hybrid sensor approach leveraged Landsat’s long-term record for trend analysis and Sentinel-2’s higher spatial resolution for monitoring smaller water bodies, balancing temporal depth and spatial detail while managing processing demands (Table 1).

Table 1 - Satellite datasets used for NDWI-based surface water analysis in northern Manitoba.

Dataset	Sentinel-2 Level-2A (S2_L2A)	Landsat 8 OLI/TIRS (C2 L2)
Provider/Source	Copernicus OpenEO Hub	USGS EarthExplorer
Temporal	June-August 2016-2021	June-August 2016-2021
Resolution	10 m	30 m
Bands/Variables	B3 (Green), B8 (NIR)	Band 3 (Green), Band 5 (NIR)
Filtering/Notes	Scenes with <20% cloud coverage; Level-2A atmospherically corrected; $NDWI = (B3 - B8) / (B3 + B8)$	Scenes with <20% cloud coverage; Level-2 Surface Reflectance Products; $NDWI = (Green - NIR) / (Green + NIR)$

3.3 Preprocessing and NDWI Computation

Imagery underwent hybrid cloud-based and local preprocessing (Figure 1). Cloud, shadow, and snow pixels were masked using metadata layers (Sentinel-2: Scene Classification Layer; Landsat: QA_PIXEL), and the imagery was clipped to the AOI to reduce irrelevant data and noise. Median temporal reduction generated annual composites, which were locally processed using histogram normalization and additional noise reduction to improve spectral consistency [13] – [15].

The NDWI was computed for each composite to enhance water-land contrast, defined as:

$$NDWI = \frac{(Green - NIR)}{(Green + NIR)}$$

where Sentinel-2 uses bands B03 (Green) and B08 (NIR), and Landsat 8/9 uses bands B03 and B05 [5]. Thresholds were applied to generate binary water masks, with values adapted to the boreal wetland landscape ($NDWI > 0.3$ for open water; $NDWI \geq 0.0$ to -0.1 for shallow or mixed water). Thresholds were refined using histogram analysis and visual inspection to balance sensitivity and minimize misclassification, following established regional practices [5]–[8], [10].

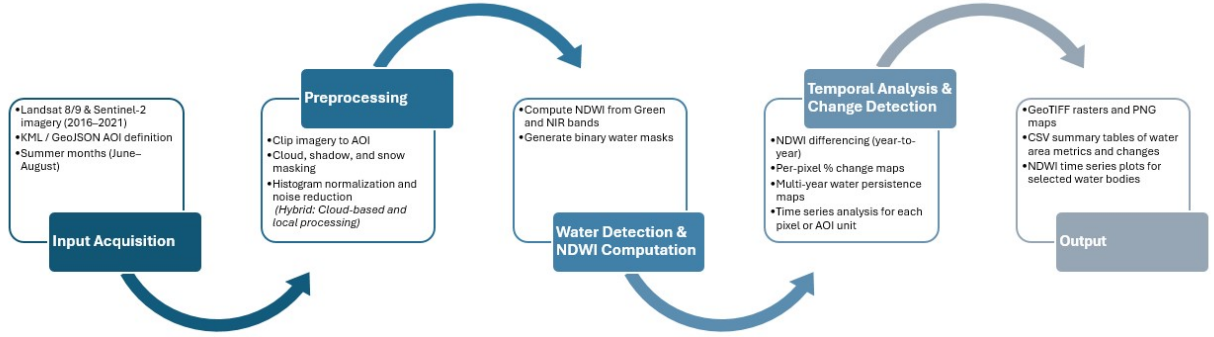


Figure 1 – Workflow from satellite imagery acquisition to NDWI computation and change detection, highlighting cloud-based and local processing steps.

3.4 Change Detection and Temporal Analysis

Annual NDWI composites were temporally aligned to quantify surface water dynamics using two complementary approaches. NDWI differencing calculated pixel-level changes between years [6], [7]:

$$\Delta NDWI = NDWI_{YearX} - NDWI_{YearY}$$

Positive values indicate water gain, and negative values indicate water loss. In parallel, binary water masks from each year were compared to produce categorical change outputs: stable water, stable land, water gain, and water loss.

Annual NDWI composites were also compiled into pixel level or aggregated AOI time series to assess long-term trends, seasonal variability, and abrupt changes in water extent, enabling both quantitative (area changes) and qualitative (persistent vs. temporary water) interpretation.

3.5 Computational Environment and Outputs

All processing was conducted on a personal workstation using Python for raster processing [13, 14] and QGIS and Google Earth Engine for visualization and AOI preparation [10, 11]. Raw imagery was stored on an external SSD, and batch processing managed memory limitations, demonstrating that multi-year surface water analysis is feasible without high-performance computing resources [10, 11]. Processed outputs included GeoTIFFs for geospatial analysis [13], PNG visualizations for interpretation, CSV summaries of water area metrics and change statistics, and NDWI time series plots for representative water bodies [5–7].

4 Results

4.1 NDWI Computation and Preprocessing

Preprocessing and NDWI computation transformed raw Sentinel-2 and Landsat-8 imagery into high-quality water maps suitable for multi-year analysis. Cloud, shadow, and snow masking, AOI clipping, median temporal reduction, and histogram normalization reduced noise and improved spectral consistency, ensuring comparability across years. Year-specific comparisons (Figure 2 and Figure 3) illustrate the RGB → NDWI → Water Mask → Preprocessed workflow, showing how water features are isolated from surrounding land cover. Multi-temporal NDWI summaries for Landsat-8 and Sentinel-2 (Figure 4 and Figure 5) highlight annual Δ NDWI and percentage change, demonstrating spatial patterns of water dynamics across 2016–2021. These figures collectively showcase the effectiveness of preprocessing and spectral

transformation in enhancing feature separation, facilitating consistent water detection, and providing a robust qualitative and quantitative foundation for subsequent change detection analyses in complex boreal and wetland landscapes.

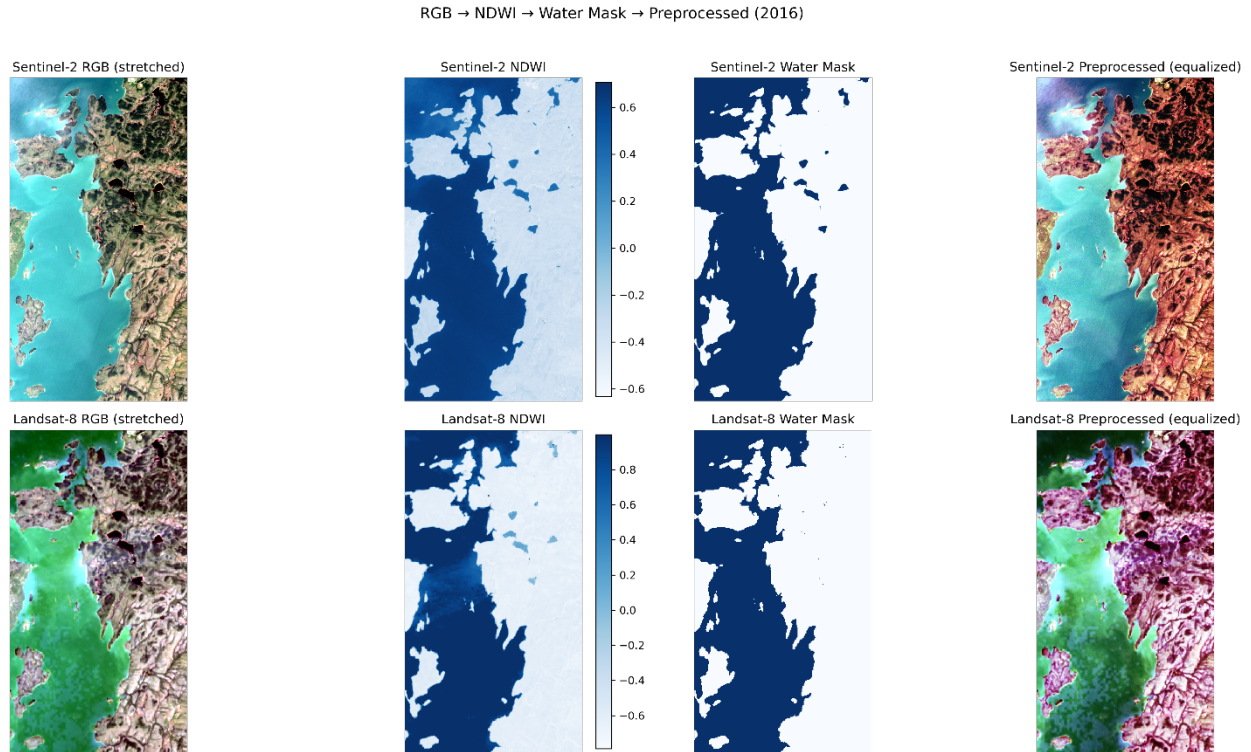


Figure 2 - RGB → NDWI → Water Mask → Preprocessed (2016): Sentinel-2 and Landsat-8 workflow for 2016. Column 1: RGB composites. Column 2: NDWI maps. Column 3: Water masks. Column 4: Preprocessed images with histogram equalization.

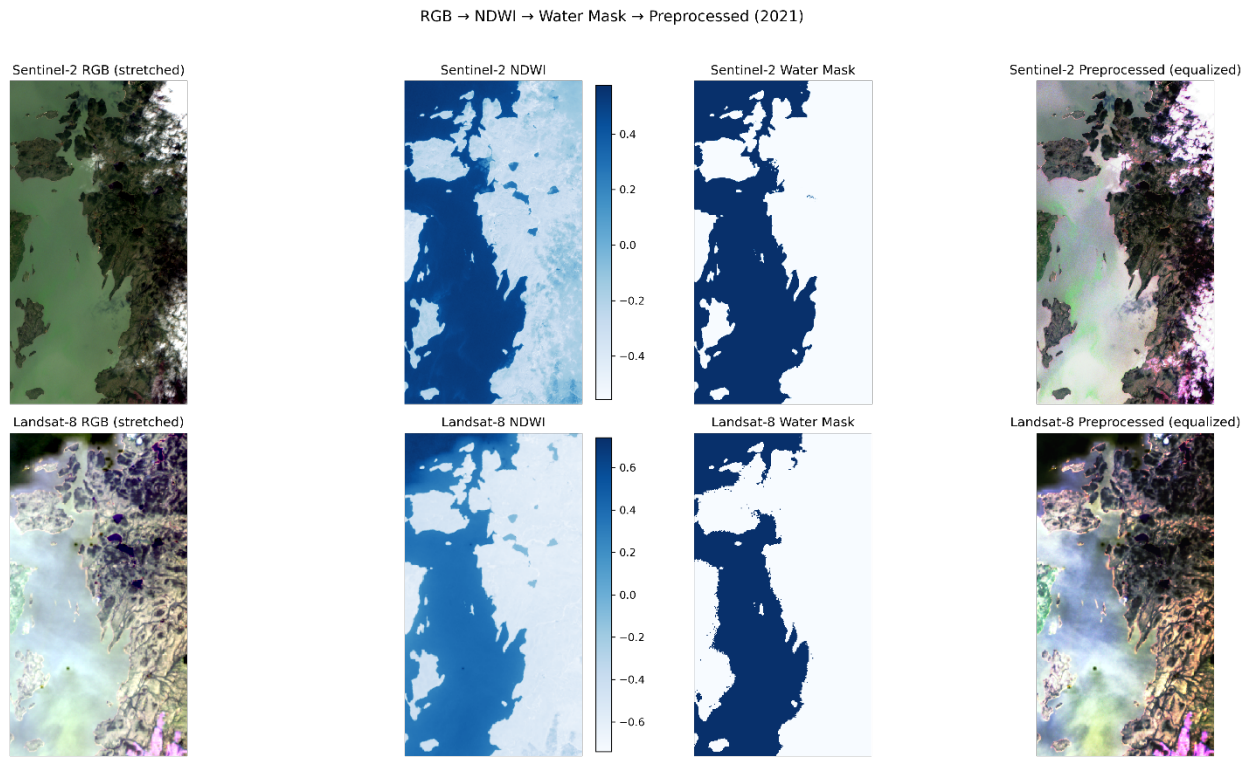


Figure 3 - RGB → NDWI → Water Mask → Preprocessed (2021): Sentinel-2 and Landsat-8 workflow for 2021. Column 1: RGB composites. Column 2: NDWI maps. Column 3: Water masks. Column 4: Preprocessed images.

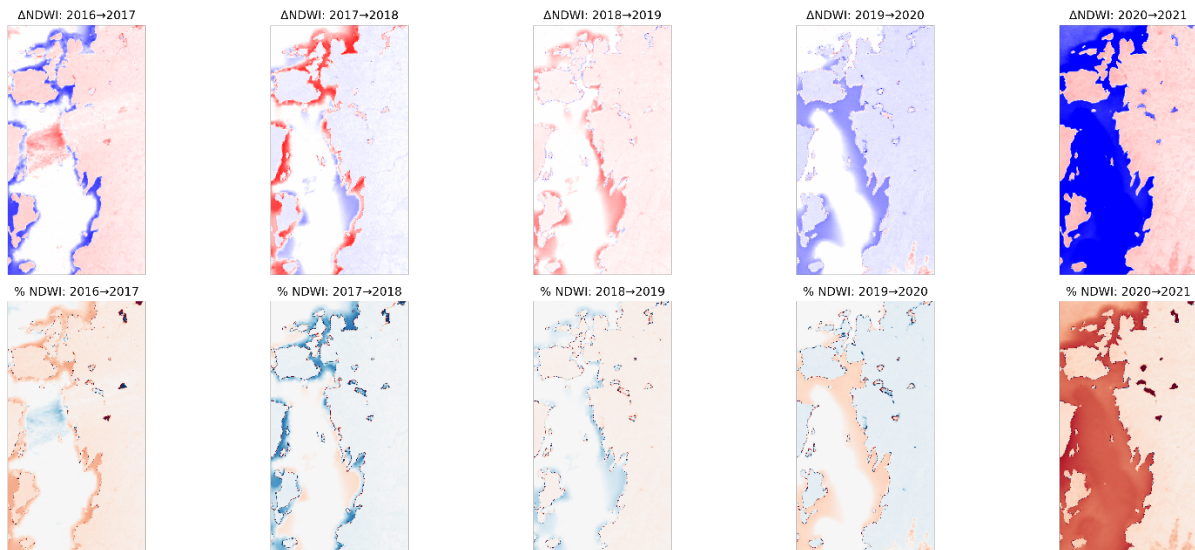


Figure 4 - Multi-temporal NDWI Change Detection (2016–2021) – Landsat-8: Annual Δ NDWI and percentage change maps showing water dynamics across the study area. Blue = increase, red = decrease.

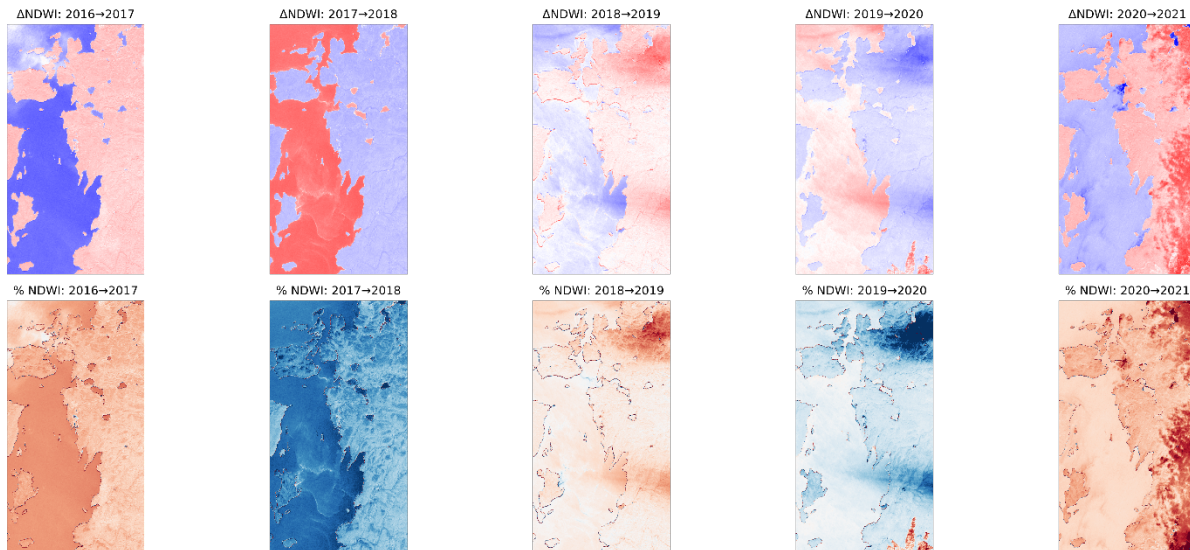


Figure 5 - Multi-temporal NDWI Change Detection (2016–2021) – Sentinel-2: Annual Δ NDWI and percentage change maps showing water dynamics. Blue = increase, red = decrease.

4.2 Change Detection Workflow

Temporal alignment of NDWI composites enabled detection of water gain, loss, and stable areas using both Δ NDWI and binary water mask comparison. Landsat-8 multi-year change and percent change maps (Figure 6 and Figure 7) reveal pixel-level water extent changes from 2016 to 2021 and normalize these differences for comparison across water bodies of varying sizes. Sentinel-2 multi-year change and percent change maps (Figure 8) show similar patterns across the study area. Multi-panel summary figures (Figure 9) consolidate these results, displaying Δ NDWI maps, binary water change masks, and NDWI with change contours, highlighting areas of persistent gain, loss, or stability. Together with Table 2, these figures illustrate how image differencing, thresholding, and multi-year aggregation produce actionable insights from multi-temporal satellite imagery while minimizing noise and spectral variability.

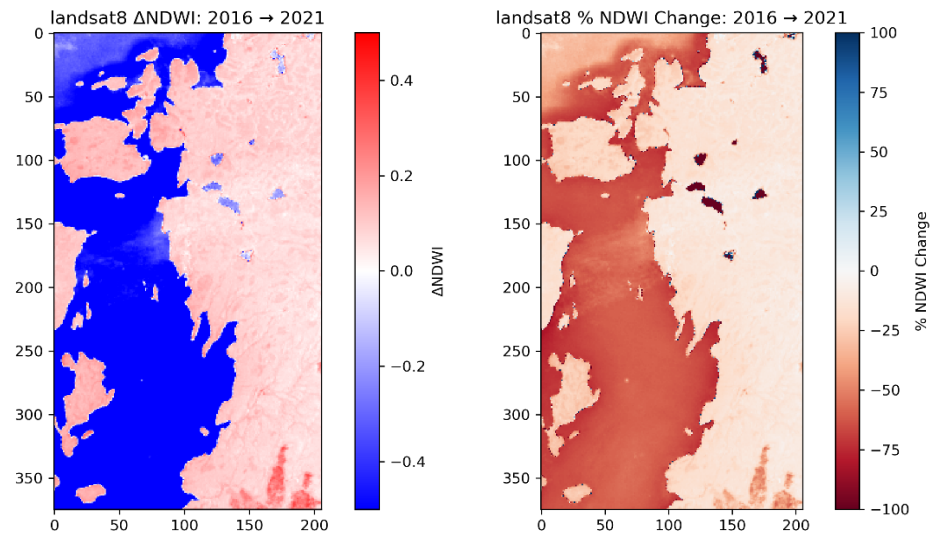


Figure 6 - Landsat-8 Multi-year NDWI Change and Percent Change (2016–2021): Left: pixel-level Δ NDWI showing water gain (blue) and loss (red). Right: percent change map normalized for comparison across water bodies of varying sizes.

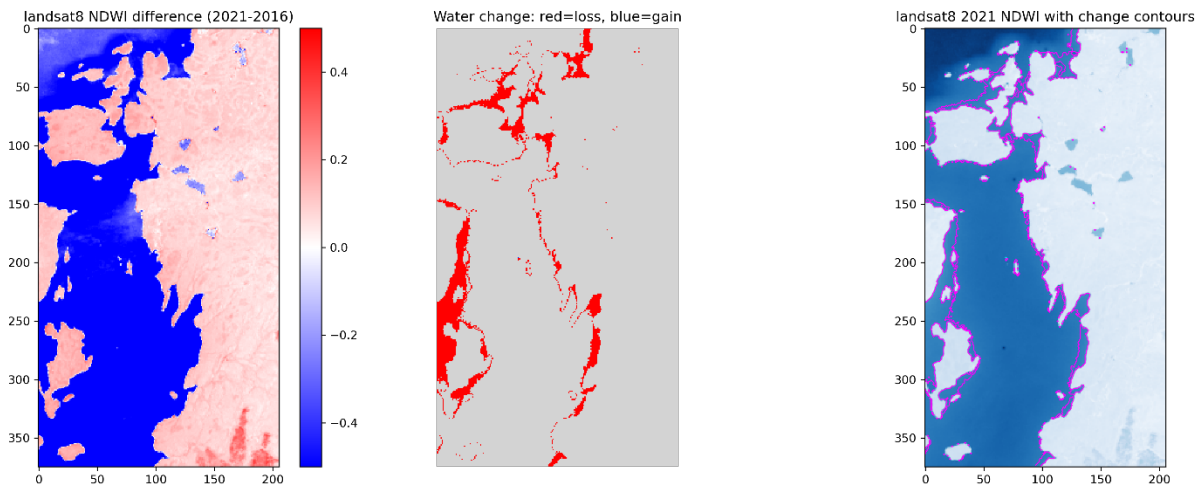


Figure 7 - Landsat-8 NDWI Change Summary: Multi-panel figure showing (1) Δ NDWI map, (2) binary water change mask, and (3) NDWI with change contours.

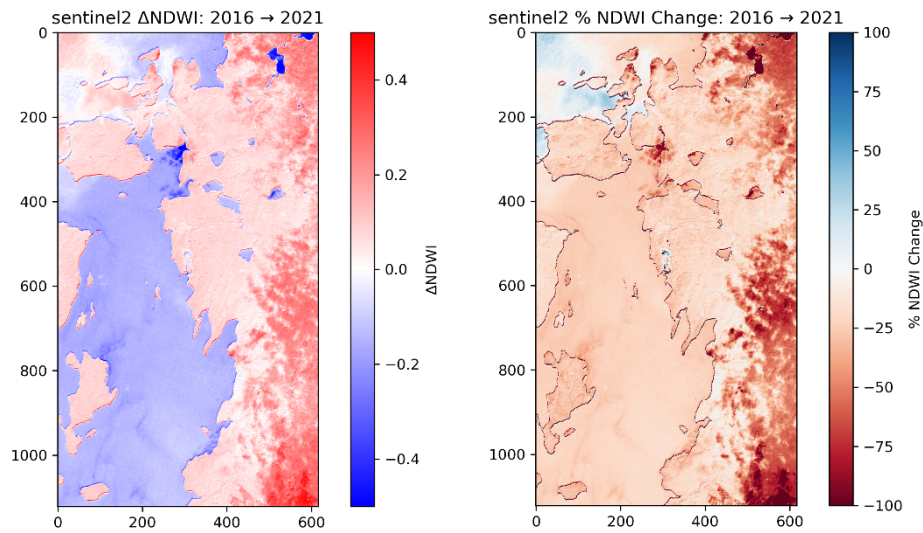


Figure 8 - Sentinel-2 Multi-year NDWI Change and Percent Change (2016–2021): Left: pixel-level Δ NDWI showing water gain and loss. Right: percent change map normalized for comparison across water bodies.

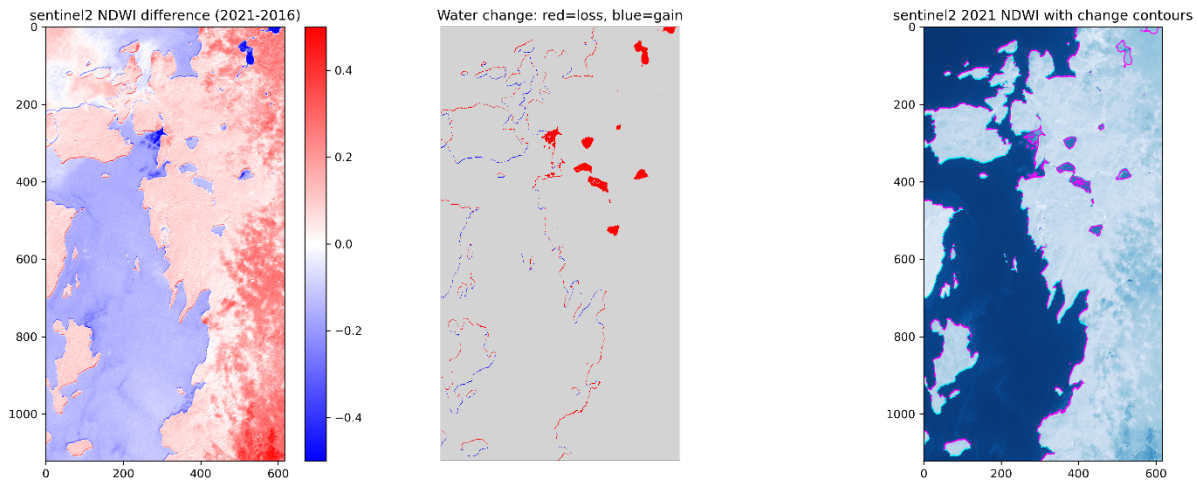


Figure 9 - Sentinel-2 NDWI Change Summary: Multi-panel figure showing (1) Δ NDWI map, (2) binary water change mask, and (3) NDWI with change contours.

Table 2 - Summary of Surface Water Change (2016–2021)

Sensor	Area Gained (km ²)	Area Lost (km ²)	Net Change (km ²)	Gain Pixels	Loss Pixels
Landsat 8	0.001	4.360	-4.359	1	4844
Sentinel-2	0.160	0.914	-0.754	1602	9140

4.3 Temporal Analysis and Water Persistence

Multi-year NDWI composites and binary water masks were analyzed to assess water persistence and temporal trends across the AOI. The persistence maps derived from Landsat 8 and Sentinel-2 imagery (Figure 10a and Figure 10b) visualize which water bodies remained stable over the five-year study period and which were temporary or seasonal. The Sentinel-2 overall NDWI composite (Figure 10c) further highlights areas of high interannual variability.

Table 3 provides annual NDWI values and corresponding water area metrics, demonstrating how temporal aggregation and pixel-level analysis of NDWI can quantify both persistent and ephemeral water features. Together, these results show how digital image processing techniques, including connected-pixel analysis, temporal filtering, and threshold-based classification, allow for the extraction of long-term trends and nuanced spatial patterns from multi-temporal satellite data.

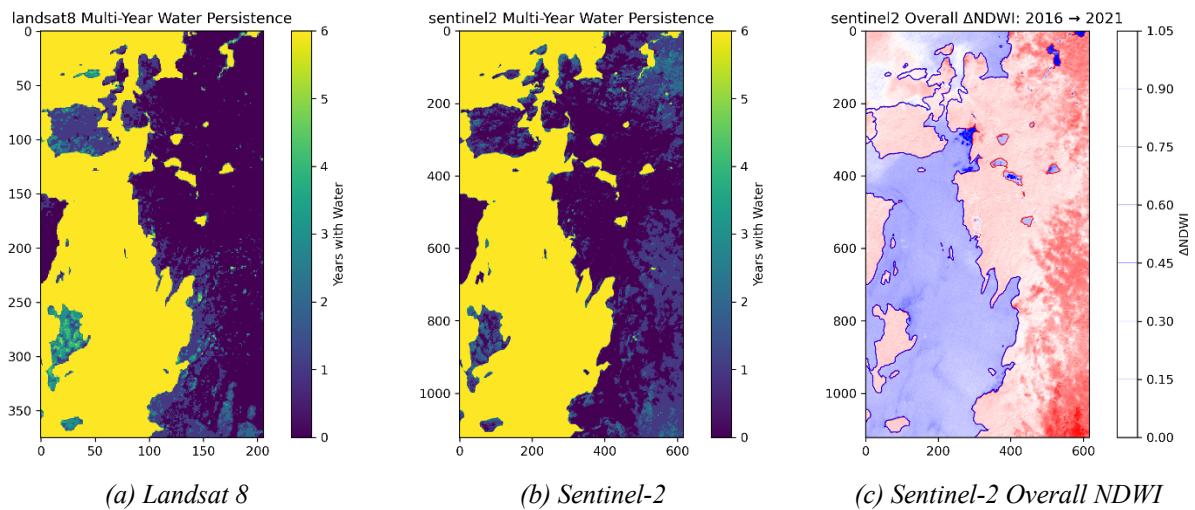


Figure 10 - Multi-temporal NDWI-derived surface water persistence and variability across the study area (2016–2021): (a) Landsat 8 water persistence, (b) Sentinel-2 water persistence, and (c) Sentinel-2 overall NDWI composite highlighting interannual variability.

Table 3 - Quantitative Surface Water Change Summary (Summer NDWI)

Sensor	Period	Water Gained (km ²)	Water Lost (km ²)	Net Change (km ²)	Gain Pixels	Loss Pixels
Landsat 8	2016–2017	23.36	9.17	+14.19	25,957	10,185
Landsat 8	2017–2018	9.94	4.87	+5.08	11,047	5,408
Landsat 8	2018–2019	9.25	1.27	+7.98	10,281	1,409
Landsat 8	2019–2020	1.38	29.06	−27.68	1,531	32,285
Landsat 8	2020–2021	35.44	30.64	+4.80	39,373	34,045
Landsat 8	2016–2021	36.59	30.18	+6.42	40,660	33,529
Sentinel-2	2016–2017	36.87	30.52	+6.35	368,683	305,172
Sentinel-2	2017–2018	30.98	37.79	−6.81	309,795	377,911
Sentinel-2	2018–2019	17.51	13.76	+3.75	175,079	137,581
Sentinel-2	2019–2020	15.10	32.58	−17.48	150,966	325,772
Sentinel-2	2020–2021	37.83	30.74	+7.08	378,270	307,444
Sentinel-2	2016–2021	33.33	26.99	+6.34	333,315	269,925

4.4 Observations

Across both sensors, large lakes remained largely stable, while smaller wetlands displayed the greatest interannual variability. The higher spatial resolution of Sentinel-2 enabled the detection of fine-scale changes that were partially obscured in Landsat imagery. The workflow demonstrates that applying preprocessing, NDWI transformation, thresholding, image differencing, and temporal aggregation provides robust identification of water changes while mitigating the influence of clouds, shadows, and mixed pixels. Visualizations of NDWI composites, difference maps, and persistence maps not only support quantitative analysis but also provide a clear illustration of the workflow’s capabilities. Although the focus here is on demonstrating digital image processing techniques, the results also allow secondary observation of the geographic distribution and temporal dynamics of surface water in northern Manitoba, confirming that the workflow can produce both operationally informative and meaningful outputs.

5 Discussion

The results demonstrate that multi-temporal satellite imagery, when processed through a robust digital image processing workflow, can effectively detect and monitor surface water changes in a resource-efficient manner. The hybrid approach, combining Landsat 8/9 and Sentinel-2 imagery, balances temporal depth and spatial resolution, allowing for both the detection of small, medium, and large water bodies and the evaluation of interannual dynamics. NDWI computation, thresholding, and image differencing proved reliable in isolating water features, even in complex boreal and wetland-dominated landscapes [5]–[8], [18].

The comparison of Landsat and Sentinel-2 imagery highlights how sensor characteristics influence detection capacity. Landsat’s moderate spatial resolution limits identification of small wetlands but provides a long-term perspective for large water bodies. In contrast, Sentinel-2’s higher spatial resolution and shorter revisit time enhance the detection of fine-scale water dynamics, emphasizing the importance of selecting sensors appropriate for the scale and temporal resolution of interest.

The workflow’s preprocessing steps, including cloud/shadow masking, AOI clipping, noise reduction, and histogram normalization, demonstrate the practical application of digital image processing techniques in producing high-quality NDWI composites. These methods effectively mitigate errors caused by atmospheric effects, mixed pixels, and residual cloud contamination, highlighting the value of Digital Image Processing (DIP) principles in environmental monitoring workflows.

6 Limitations and Future Work

This study has several limitations that suggest directions for future research. First, the AOI was limited to 69.23 km² due to computational constraints, restricting geographic

generalizability [9–11]; future work could extend to larger AOIs and longer temporal ranges using cloud-based high-performance computing [11, 12]. Second, while NDWI effectively detects water, it may misclassify shallow water, saturated soils, or dense vegetation, particularly with uniform thresholds [5–7]. Using additional indices such as MNDWI or AWEI could improve classification, especially for shallow or turbid waters [6, 8]. Third, the workflow depends on atmospherically corrected Level-2 imagery, so errors propagate through NDWI and change detection [9, 10]; integrating machine learning could reduce misclassification of mixed or ephemeral water bodies [8]. Finally, analyses on a personal workstation constrained spatial and temporal extent [9]. Combining satellite workflows with field observations would allow quantitative validation and increase confidence in DIP-based monitoring [16, 18].

7 Conclusion

This study presents a proof-of-concept workflow for detecting and monitoring surface water changes using multi-temporal satellite imagery in northern Manitoba. By integrating DIP techniques such as cloud/shadow masking, AOI clipping, histogram normalization, NDWI computation, thresholding, and image differencing, the workflow produces reliable NDWI composites, binary water masks, and change maps. The hybrid use of Landsat 8/9 and Sentinel-2 imagery demonstrates how sensor-specific characteristics influence detection capacity, with Sentinel-2 capturing fine-scale water dynamics and Landsat providing temporal depth.

Despite hardware and spatial limitations, the workflow demonstrates that meaningful multi-year surface water monitoring can be conducted using standard computing resources. These findings emphasize the applicability of digital image processing principles in environmental monitoring and provide a scalable methodology for future studies of surface water dynamics in boreal and sub-Arctic regions.

8 References

- [1] W. L. Quinton, M. Hayashi, and L. Chasmer, “Permafrost–groundwater interactions and hydrologic connectivity in the subarctic,” *Hydrol. Processes*, vol. 25, no. 21, pp. 3476–3487, 2011.
- [2] T. Prowse et al., “Arctic freshwater synthesis: Summary of key emerging issues,” *J. Geophys. Res.: Biogeosciences*, vol. 120, no. 10, pp. 1887–1893, 2015.
- [3] S. L. Smith, “Thermal state of permafrost in North America,” *Permafrost Periglacial Processes*, vol. 21, no. 2, pp. 117–135, 2010.
- [4] J. Kokelj, “Thaw-driven mass wasting couples slopes with downstream systems,” *J. Geophys. Res.: Earth Surface*, vol. 118, no. 2, pp. 1106–1120, 2013.
- [5] S. K. McFeeters, “The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features,” *Int. J. Remote Sensing*, vol. 17, no. 7, pp. 1425–1432, 1996.
- [6] H. Xu, “Modification of Normalized Difference Water Index (NDWI) to enhance open water features,” *Int. J. Remote Sensing*, vol. 27, no. 14, pp. 3025–3033, 2006.
- [7] L. Ji, L. Zhang, and B. Wylie, “Analysis of dynamic thresholds for the Normalized Difference Water Index,” *Photogramm. Eng. Remote Sensing*, vol. 75, no. 11, pp. 1307–1317, 2009.
- [8] F. Feyisa, “Automated Water Extraction Index: A new technique for surface water mapping,” *Remote Sens. Environ.*, vol. 140, pp. 23–35, 2014.
- [9] U.S. Geological Survey, *Landsat Level-3 Dynamic Surface Water Extent, Collection 2*, 2022.
[Online]. Available: <https://doi.org/10.5066/P9DPWBUS>
- [10] European Space Agency, *Copernicus Sentinel-2 Satellite Imagery*. [Online]. Available: <https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-data/sentinel-2>

- [11] N. Gorelick, “Google Earth Engine: Planetary-scale geospatial analysis for everyone,” *Remote Sens. Environ.*, vol. 202, pp. 18–27, 2017.
- [12] openEO Foundation, *openEO API for Earth Observation*. [Online]. Available: <https://openeo.org>
- [13] Rasterio Development Team, *Rasterio: Access to Geospatial Raster Data*. [Online]. Available: <https://rasterio.readthedocs.io/en/latest/>
- [14] OpenCV Development Team, *Reading Geospatial Raster Files with GDAL*. [Online]. Available: https://docs.opencv.org/4.x/d7/d73/tutorial_raster_io_gdal.html
- [15] Sentinel Hub, *Custom Scripts Repository*. [Online]. Available: <https://custom-scripts.sentinel-hub.com/custom-scripts/>
- [16] D. Atkinson, “Participatory mapping, Indigenous knowledge, and Arctic environmental change,” *Appl. Geography*, vol. 54, pp. 102–112, 2014.
- [18] R. Rockwell, “Landscape change and climate gradients across northern Manitoba,” *Can. J. Remote Sensing*, vol. 47, no. 1, pp. 85–101, 2021.